Final Group Project Report: ScamShield US

Generative AI & Natural Language Processing — Agentic RAG for Consumer Fraud Education

1. Introduction

This report presents **ScamShield US**, an open-source agentic retrieval-augmented generation (RAG) system for consumer fraud and identity-theft education. A user submits a suspicious message; the system classifies risk with Traditional / Advanced / Hybrid NLP models, retrieves the top five official guidance passages, and generates an educational answer with a local Qwen decoder. Durable artifacts are stored in Amazon S3 and accessed only through boto3; compute runs on AWS EC2. The remainder of this report describes data, models, experimental setup, hyperparameters, results, conclusions, and references.

2. Description of the data set

2.1 Classification corpora

We combine the UCI SMS Spam Collection (`sms_spam.csv`, Category/Text) and the Super SMS Dataset (`super_sms_dataset.csv`, SMSes/Labels). Files are stored at `s3://…/nlp-data/data/raw/`. After merge/dedup, processed splits are written to `data/processed/` (messages_train/val/test/all.csv). Binary ham/spam labels come from the sources; finer scam subtypes used in multi-class experiments are silver/heuristic labels where applied.

2.2 Knowledge corpus for RAG

We seed 24 URL-backed official consumer-protection documents (FTC, CISA, FCC, SSA, IRS, SEC, etc.) into `data/knowledge/` with a provenance inventory. These documents are chunked and indexed for hybrid retrieval.

3. NLP models and algorithms

3.1 Traditional NLP

Messages are converted to TF-IDF features (with domain stopwords) plus lightweight linguistic/scam-signal features. We train Logistic Regression and a calibrated Linear SVM, then majority-vote for the Traditional prediction. This corresponds to classical/rule-oriented discriminative models from the course concept list.

3.2 Advanced NLP (pretrained transformer)

We fine-tune DistilBERT (`distilbert-base-uncased`) as a sequence classifier. DistilBERT uses multi-head self-attention layers distilled from BERT, providing contextual token representations for scam-type discrimination.

3.3 Hybrid cascade

Hybrid NLP keeps the Traditional label when the Traditional confidence margin is strong; otherwise it escalates to DistilBERT. The margin is selected on the validation split (final choice: 0.05).

3.4 Hybrid RAG + decoder LLM

Retrieval uses BM25 (sparse) and FAISS over MiniLM embeddings (dense), fused into a hybrid ranking with **top_k = 5**. The answer model is Qwen2.5-1.5B-Instruct, a decoder-only causal language model that predicts next tokens under a causal attention mask. The prompt contains only retrieved official passages plus the user question; generation is greedy (`do_sample=False`).

4. Experimental setup

Data are split into train/validation/test with a fixed seed. Models are trained on train, Hybrid margin is chosen on validation, and final comparison tables are computed on test by `scripts/evaluate_approaches.py`. Implementation uses PyTorch + Hugging Face Transformers on EC2 GPU, scikit-learn for classical models, and FastAPI/Angular for serving. Performance for classifiers is judged primarily by macro-F1, with accuracy and weighted-F1 as secondary metrics. Artifacts are uploaded to S3 via boto3.

5. Hyperparameters and overfitting controls

- DistilBERT sequence length capped (e.g., 160 tokens in evaluation scoring); AdamW via Transformers Trainer defaults unless overridden in training script.
- Hybrid margins searched on validation: {0.05, 0.10, 0.15, 0.20, 0.30}; selected 0.05.
- RAG: chunk size/overlap and hybrid alpha configured in `configs/settings.yaml`; final context limited to top_k=5.
- Qwen: max_new_tokens=320, temperature effectively greedy (do_sample=False).
- Overfitting controls: held-out test never used for model selection; validation for Hybrid margin; class_weight='balanced' for classical models; early stopping / eval on val during DistilBERT training where configured.

6. Results

Table 1 reports held-out test metrics from `approach_metrics.csv`. Hybrid NLP attains the best macro-F1 (0.983) and accuracy (0.992), while routing only ~1.1% of test messages to DistilBERT. Traditional and Advanced NLP are both strong baselines; the cascade improves rare-class balance (macro-F1) while keeping most traffic on the cheaper Traditional model. Dashboard wordclouds are built from labeled train spam/ham term frequencies (not model attention weights).

Approach	Macro-F1	Accuracy	Weighted-F1	Routed to Adv.
Traditional NLP	0.969	0.987	0.987	0%
Advanced NLP	0.961	0.987	0.987	100%
Hybrid NLP	**0.983**	**0.992**	**0.992**	**1.1%**

Table 1. Test-set comparison of classification approaches.

Figure 1. Term-frequency wordclouds from labeled training spam (left) and ham (right). These are corpus statistics for the dashboard, not DistilBERT attention maps.



The Angular UI also loads `approach_reports.json` for confusion matrices and per-label metrics from the same evaluation run.

7. Summary and conclusions

We built an end-to-end educational fraud assistant combining classical NLP, DistilBERT, hybrid retrieval, and a local Qwen decoder under EC2+S3 constraints. Hybrid cascading improved macro-F1 while keeping most predictions on the cheaper Traditional model. Future improvements include stronger interpretability (LIME/SHAP/IG), optional RNN baselines for comparison, larger knowledge coverage, and higher-capacity open LLMs with quantization on A10G GPUs.

8. References

- UCI SMS Spam Collection — https://archive.ics.uci.edu/dataset/228/sms+spam+collection
- Super SMS Dataset / spstudy — https://github.com/smspamresearch/spstudy
- DistilBERT — Sanh et al., 2019; Hugging Face Transformers
- Sentence-Transformers MiniLM — Reimers & Gurevych
- Qwen2.5 — Alibaba Qwen team (Hugging Face model card)
- FAISS — Johnson et al.; Facebook AI Similarity Search
- Project repository — https://github.com/DhanushGWU1995/sms-spam-detection
- Course constraint materials / boto3 AWS SDK documentation